Thus the selected double glazing saves the following cost over 12 :

$$\triangle C_{\min} = C_{\min} - C'_{\min} = 21.47 \text{ yuan.} \qquad (19)$$

4.

1. First let one electron tunnel from plate A to B. Let Q be the charge on plate A before tunnelling, and V_{AB} the plate voltage (problem Fig. 3). Then

$$V_{AB} = Q / C \ . \qquad (1)$$

The stored capacitor energy is

$$U = \frac{1}{2} C V^2_{AB} \ . \qquad (2)$$

After the electron tunnels to B, the charge on A is

$$Q' = Q + e \ , \qquad (3)$$

Here e is electron charge magnitude. The new voltage and stored energy are

$$V'_{AB} = \frac{Q + e}{C} \ , \ U' = \frac{1}{2} C V'^2_{AB} \ . \qquad (4)$$

Coulomb blockade, forbidding tunnelling, requires

$$U' - U > 0 \ . \qquad (5)$$

Equations (1) ~ (5) give

$$V_{AB} > -\frac{1}{2}\frac{e}{C} \ . \qquad (6)$$

Now let an electron tunnel from B to A. Again Q is the charge on A before tunnelling, V_{AB} the plate voltage. When it tunnels from B to A, plate A has charge $Q' = Q - e$. Similarly, forbidding tunnelling from B to A requires the following condition:

$$V_{AB} < \frac{1}{2}\frac{e}{C} \ . \qquad (7)$$

From (6), (7), voltage V_{AB} between $-e/2C \sim e/2C$ blocks single-electron tunnelling. The blockade condition is

$$-\frac{1}{2}\frac{e}{C} < V_{AB} < \frac{1}{2}\frac{e}{C} \ . \qquad (8)$$

2. By the stated condition and (8), the tunnelling threshold satisfies

$$V_{AB} = \frac{1}{2}\frac{e}{C} = 0.10 \text{ mV} \ . \qquad (9)$$

Substituting the data into (9) gives

$$C = 8.0 \times 10^{-16} \text{ F} \ . \qquad (10)$$